

CME Review

Current concepts: Common variable immunodeficiency



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Key Messages

- Common variable immunodeficiency is the most frequent symptomatic primary immunodeficiency disorder found in clinical medicine.
- It is characterized by lower than normal IgG, IgA, and/or IgM deficiency with proven loss of functional antibody.
- Patients present with infections in most cases, but some have autoimmune or inflammatory conditions as well.
- Genetic defects can be identified currently in approximately 30% of subjects, regardless of age.
- Immunoglobulin replacement is needed in all subjects.
- Biologics can provide benefit for those with autoimmune or inflammatory conditions, but further treatment approaches are needed.

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ABSTRACT

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Common variable immunodeficiency (CVID) is the most frequent symptomatic primary immunodeficiency disorder found in clinical medicine. The immune defect occurs in males and females equally and is characterized by reduced serum levels of IgG, deficient IgA and/or IgM levels, and poor to absent specific antibody responses to infection or vaccinations. Although CVID is considered a primary immune defect, most subjects are diagnosed between the ages of 20 and 40 years old. However, due to the heterogeneous clinical appearance, a diagnostic delay of 5 to 8 years after the first cardinal symptom is common in all countries in which this has been investigated. As the genetics of this immune defect have been further clarified, it is clear that the name “CVID” is an umbrella diagnosis, useful clinically for arranging treatment, but it actually includes a very large number of immune defects, many of which are not yet discovered. Due to prevalence, common inflammatory complications, and numbers of medical encounters, the awareness of CVID is critical for pediatricians, internists, and primary care physicians, including pulmonologists, otolaryngologists, hematologists, and physicians in many other specialties. Here, we outline the early history of CVID, the diagnostic criteria and standard workup of subjects, the clinical manifestations, emerging genetic understandings, and current treatment modalities for patients with this immune defect.

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Introduction

Common variable immunodeficiency (CVID) is the most frequent symptomatic primary immunodeficiency disorder found in clinical medicine. The immune defect is characterized by reduced serum levels of IgG, deficient IgA and/or IgM, and poor to absent specific antibody responses to infection or vaccinations.^{1–3} The incidence of

CVID is not actually known, but it has been estimated at 1:20,000 and 1:50,000 in Whites in the United States, and there are clearly regional differences, with CVID being apparently less often diagnosed in Asians and African Americans.⁴ Although CVID is considered a primary immune defect, most subjects are diagnosed between the ages of 20 and 40 years old. However, due to the heterogeneous clinical appearance, a diagnostic delay of 5 to 8 years after the first cardinal symptom is common in all countries in which this has been investigated.⁵ As the genetics of this immune defect have been further clarified, it is clear that the name “CVID” is an umbrella diagnosis, useful clinically for arranging treatment, but it actually includes a very large number of immunologic mechanisms leading to loss of antibody

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Participants will be able to demonstrate increased knowledge of the clinical treatment of allergy/asthma/immunology and be able to apply new information to their own practices.

Learning Objectives

At the conclusion of this activity, participants should be able to:

- Summarize the clinical presentation of common variable immune deficiency (CVID).
- Specify the diagnostic tests that are best used to evaluate a patient with suspected common variable immune deficiency.

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Target Audience

Physicians involved in providing patient care in the field of allergy/asthma/immunology.

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production, many of which are not yet discovered. Due to prevalence and numbers of medical encounters, the awareness of CVID is critical for pediatricians, internists, and primary care physicians, including pulmonologist, otolaryngologists, hematologists, and physicians in many other specialties.^{4,6,7} In this review, we outline the early history of CVID, the diagnostic criteria and standard workup of subjects, the clinical manifestations, emerging genetic understandings, and current treatment modalities for patients with this immune defect.

Early History

After the first report of a boy with agammaglobulinemia appeared in 1952,⁸ other medical centers began to identify similar males in their own patient cohorts, medical events that were in many ways the initiation of the field of primary immune deficiency. However, the report of a 39-year-old woman 2 years later with agammaglobulinemia, who had experienced recurrent episodes of bacterial pneumonia, *Haemophilus influenzae* meningitis, chronic diarrhea with malabsorption, and severe weight loss,⁹ was a medical surprise, as of course this could not be X-linked, and for a genetic defect, the patient was considered too old. However, in the next several decades, a number of other reports described the development of bronchiectasis and respiratory failure (and 1 with lymphoma)¹⁰ in adults with agammaglobulinemia or hypogammaglobulinemia. Early investigators were struck by the observation that such patients often “present a history indicating that they have been well and without inordinate trouble from infections up to a certain point in their lives, then suddenly, just as with congenital agammaglobulinemia, life for them becomes a continuous round of severe, life-threatening bacterial infections.”¹¹ These reports of adults suggested the need to consider this as an important immune defect, but the cause was difficult to explain. By 1970, a group of mostly pediatricians met under the auspices of the World Health Organization to classify the increasing reported “primary immune defects, accruing to the pathology encountered.” To include the all subjects with agammaglobulinemia or hypogammaglobulinemia, regardless of age, a final category was invented: “Variable Immune deficiency (common, largely unclassified).” This

category, in which the diagnosis is still known as “common variable immune deficiency” (or CVID), is still in use today.

Definition, Laboratory Evaluations

CVID is both clinically and immunologically heterogeneous, and the term is used clinically to identify subjects who have B cells but a variable lack of immune globulins IgG, IgA, and or IgM. From an immunologic point of view, this syndrome overlaps with other B disorders such as IgA deficiency, IgG and IgG subclass deficiency, hyper IgM syndromes, and antibody deficiency (Fig 1). However, diagnostic definitions are needed in medical practice, and for CVID, these have been outlined in various resources.^{1,2} (Table 1A and B). The main diagnostic criterion is the reduction of serum IgG by 2 or more SDs below the normal range for age, along with at least 1 of the other 2 isotypes (IgA or IgM). Almost all patients with CVID are IgA deficient, whereas levels of IgM are usually lower than normal, but this is more variable. An essential part of this definition is that there is a loss of specific antibody responses. This is demonstrated by the absence of protective responses to T-dependent vaccines, such as tetanus and diphtheria, and/or other vaccines, such as measles, mumps, and rubella.^{12,13} Loss of antibody to T-independent antigens (polysaccharide responses/isoheamagglutinins) should also be documented. For this, antibody responses to nonconjugated pneumococcal polysaccharide vaccine are generally tested, comparing levels before and 4 weeks after vaccination. This may also be required for ordering immunoglobulin replacement. Extensive specific antibody testing for subjects with very low serum IgG levels (200 mg/dL or less) is not really necessary because a severe humoral immune deficiency has been demonstrated and the probability of protective antibody responses is quite low.¹ However, for ordering immune globulin treatment, it may still be required.

The inability to produce immunoglobulin is a reflection of the deficiency of circulating isotype-switched memory B cells (CD27+ IgM– IgD– cells) found in many patients.^{14–17} The defects of B-cell differentiation also result in the loss of plasma cells in lymph nodes, gastrointestinal tissues, and bone marrow.¹⁸ Examination of B-cell phenotypes for the numbers of isotype-switched memory B cells has proved useful clinically, as those with very low numbers, less than 2% in 1 study¹⁷

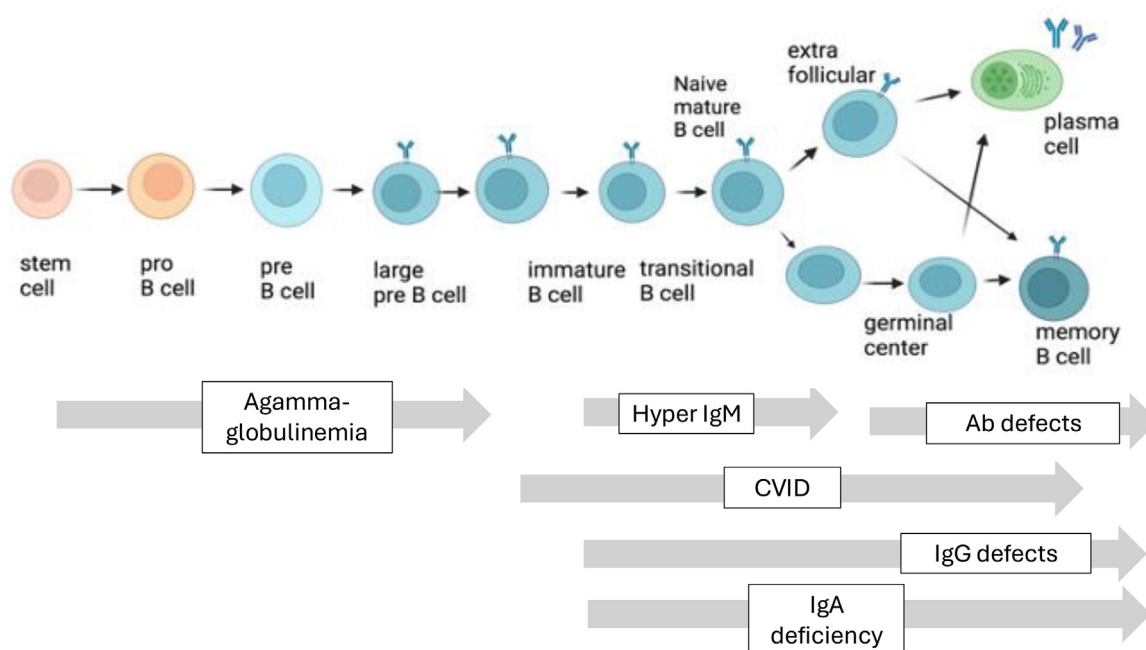


Figure 1. Scheme of B-cell development with the primary immune defects overlapping in phenotype. CVID, common variable immunodeficiency.

Table 1A
Common Variable Immunodeficiency: Diagnostic Criteria*

Probable:
Male or female patient who has a marked decrease of IgG (at least 2 SD below the mean for age) and a marked decrease in either IgM or IgA and fulfils all the following criteria:
1) Onset of immune deficiency at >2 y of age
2) Absent isohemagglutinins and/or poor response to vaccines
3) Defined causes of hypogammaglobulinemia have been excluded (Table 2)
Possible:
Male or female patient who has a marked decrease (at least 2 SD below the mean for age) in at least 1 of the major isotypes (IgM, IgG, and IgA) and fulfils all the following criteria:
1) Onset of immune deficiency at >2 y of age
2) Absent isohemagglutinins and/or poor response to vaccines
3) Defined causes of hypogammaglobulinemia have been excluded (Table 2)

NOTE. * Adapted from Bonilla et al,¹ 2016.

Table 1B
Common Variable Immunodeficiency: European Society for Immunodeficiencies Diagnostic Criteria*

1. At least 1 of the following:
• Increased susceptibility to infection
• Autoimmune manifestations
• Granulomatous disease
• Unexplained polyclonal lymphoproliferation
• Affected family member with antibody deficiency
2. AND
• Marked decrease of IgG and marked decrease of IgA with or without low IgM levels
• AND at least 1 of the following:
• Poor antibody response to vaccines (and/or absent isohemagglutinins)
• Low switched memory B cells (<70% of age-related normal values)
3. AND secondary causes of hypogammaglobulinemia have been excluded.
4. AND
• Diagnosis is established after the fourth year of life (but symptoms may be present before)
5. AND no evidence of profound T-cell deficiency for age or absent T-cell proliferation

NOTE. * Adapted from Seidel et al,² 2019.

and 0.55% in another,¹⁹ are likely to have more autoimmune/inflammatory complications.^{17,19} Additional B-cell markers include increased CD21^{low} and transitional B cells noted in subjects with autoimmune/inflammatory disease, but for clinical diagnosis, these markers are not required. For T cells, the percentage of naive CD45RA+CD4 T cells is also an informative parameter,^{20,21} because reduction of these cells is often associated with a more complex phenotype. Very low CD4 T cells (<200/ μ L) or very low numbers or percentage of naive T cells suggest more of a combined defect which may require additional disease management. Other informative clues from the initial medical evaluation of patients with CVID come from the baseline level of serum IgM; those subjects with lowest IgM levels are more likely to develop bronchiectasis,²² whereas those subjects with retained IgM are more likely to have lymphoid interstitial lung disease,²³ hepatomegaly, reduced survival, and/or develop lymphoma.^{24,25}

Genetic Considerations

CVID is considered a primary immune deficient, but only in the past 2 decades have some of the genetic causes been identified.^{3,26} Currently, approximately 30% of patients can be found to have a genetic mutation, regardless of age.^{27,28} In fact, genetic work in CVID in the past 20 years has slowly revealed some of the complex genetic machinery needed to activate and sustain long-term antibody functions in healthy humans. Some of these genes are intrinsic B-cell genes that are needed for activation or cell proliferation (CD19, CD20, CD81, CD21, BAFF receptor, TACI, IKAROS, etc.), whereas others are more broadly expressed but required for normal B-cell function (NFKB1, NFKB2, PIK3CD, PIK3R1, etc.). Other genes identified in large patient groups are those that are generally associated with

agammaglobulinemia (TCF3). Curiously, some of the genes that are identified in subjects with the clinical phenotype of “CVID” are classified as defects in other immunologic controls,³ such as control of regulatory T cells (CTLA4, LRBA, STAT3, and BACH2). Other identified genes may be associated with combined immune defects (RAG1, RAG2, ICOS, and ICOS-L) or defects of DNA repair (LIG1). Occasionally an X-linked gene (BTK, CD40L, and WAS) or a mutation linked to a genetic syndrome, previously missed in the patient, will be identified as the real cause of antibody deficiency (DiGeorge or Kabuki syndrome)^{27,29,30} (Table 2). The mutations that lead to the “CVID” clinical phenotype may be autosomal dominant or recessive (or even X-linked), but the distribution of these is not the same in all countries.²⁷ In the United States, the genetic changes identified in patients with CVID are more likely to be autosomal dominant in inheritance, but as these gene defects are variably expressed, for many patients there are no obviously affected relatives. However, the same autosomal dominant mutation can be found in apparently healthy, first-degree relatives, again revealing the variable penetrance of some of these genes. Although the genetic causes of CVID can be found in patients of all ages and clinical phenotypes, in general, the clinical phenotype is also not well correlated to the genetic basis of the immune defect.³¹ However, having the genetic background of the patient can be important, not only for understanding the inheritance patterns or family planning, but possibly also for prognosis or long-term monitoring. In some cases, the gene defect may suggest additional medical treatment approaches for inflammatory disease manifestations that result from selected gene defects (such as abatacept for subjects with CTLA4 or LRBA gene mutations³² or leniolisib for subjects with mutations in PIK3CD or PIK3R1).³³

Clinical Presentations

The diagnosis of CVID is generally made on the basis of 1 of 2 main clinical presentations, as pointed out almost 20 years ago, due to either an acute bacterial infection or a series of chronic, often respiratory infections, or alternatively, 1 or more of a group of autoimmune or inflammatory complications, with or without significant infections.⁶ In the general medical office, most patients will have had infections, whereas for specialty medical centers, due to referral patterns, 50% or more of the patients with CVID seen will have had more of the autoimmune/inflammatory conditions.

Acute or Chronic Infections

With lack of antibody, one of the most common first manifestations of CVID is a severe bacterial pneumonia, often due to *Streptococcus pneumoniae*, *H influenzae*, or a mycoplasma species.^{7,34–38} (Fig 2A–D). Other severe bacterial infections such as sepsis, meningitis, or osteomyelitis, often with the same group of organisms, are less common but are noted in all series of newly diagnosed patients. Hard to diagnose mycoplasma infections can be quite severe and may involve the lungs, urinary tract, or joints, leading to joint destruction. In other patients, a series of upper or lower respiratory tract infections have occurred, overlapping with what is assumed to be “allergies.” More rarely, an unusual infection, such as mumps in a previously vaccinated patient, or shingles at an early age may appear. Opportunistic infections (*Pneumocystis carinii*, *Mycobacterium avium*, fungal infections, cytomegalovirus, etc.) are not at all common in CVID, unless some form of immune suppression has been used. If these are found in the absence of immune suppression, the patient is more likely to have a combined immune deficiency with low T-cell numbers, and further evaluation for cellular immunodeficiencies should be considered.^{38–40}

Other common infectious presentations of CVID are gastrointestinal infections, leading to transient or persistent diarrhea.^{41–43} *Giardia lamblia* has been a common organism and should trigger evaluation for antibody deficiency if not already known. *Salmonella* species, *Campylobacter jejuni*, and *Yersinia* infections also occur.^{43,44} Norovirus has

Table 2
Commoner Genetic Mutations in Common Variable Immunodeficiency

Gene variants	Name	Inheritance
<i>TNFRSF13B</i>	Transmembrane Activator and CAML Interactor	AD
<i>NFKB1</i>	Nuclear Factor Kappa B Subunit 1	AD
<i>NFKB2</i>	Nuclear Factor Kappa B Subunit 2	AD
<i>IRF2BP2</i>	Interferon regulatory factor-2 binding protein	AD
<i>CTLA4</i>	Cytotoxic T-Lymphocyte Associated Protein 4	AD
<i>IKZF1</i>	IKAROS Family Zinc Finger 1	AD
<i>TCF3</i>	Transcription Factor 3	AD
<i>BACH2</i>	BTB Domain and CNC Homolog 2	AD
<i>KMT2D</i>	Lysine Methyltransferase 2	AD
<i>STAT3</i>	Signal Transducer And Activator Of Transcription 3	AD
<i>PIK3CD</i>	Phosphatidylinositol-4,5-Bisphosphate 3-Kinase Catalytic Subunit Delta	AD
<i>LRBA</i>	LPS Responsive Beige-Like Anchor Protein	AR
<i>CXCR4</i>	C-X-C Motif Chemokine Receptor 4	AD
DiGeorge 22q11 or <i>TBX1</i>	DiGeorge syndrome	AD
<i>WAS</i>	Wiskott Aldrich syndrome	XL
<i>RAG1/RAG2</i>	Recombination Activating Genes 1 / 2	AR
<i>AICDA</i>	Activation induced cytidine deaminase	AR
<i>STXBP2</i>	Syntaxin Binding Protein 2	AR
<i>PMM2</i>	Phosphomannomutase	AR
<i>PIK3R1</i>	Phosphoinositide-3-Kinase Regulatory Subunit 1	AD/AR
<i>LIG4</i>	DNA Ligase 4	AR
<i>JAK1</i>	Janus Kinase 1	AR
<i>IKBKG</i>	Regulatory gamma subunit of the IKB kinase (IKK)	XL
<i>TBX1</i>	T-box protein 1.	AD
<i>PMS2</i>	PMS1 Homolog 2, Mismatch Repair System	AR
<i>FOXP3</i>	Fork-winged helix family	XL
<i>LIG1</i>	DNA Ligase 1	AR
<i>CIITA</i>	Master regulator of MCH class II gene transcription	AR
<i>BTk</i>	Bruton Tyrosine Kinase	XL
<i>ADA2</i>	Adenosine deaminase 2	AR
<i>CD40L</i>	CD40 Ligand	XL
<i>RTEL1</i>	regulator of telomere elongation helicase 1	AR

AD = autosomal dominant AR = autosomal recessive XL = X- linked.

become a recently noted organism and can be a cause of quite severe, hard-to-treat chronic gastrointestinal disease.⁴⁵ *Cryptosporidium parvum* and cytomegalovirus infections have been described but need to be considered as signs of a more complex form of immune deficiency. In addition, patients with CVID with gastrointestinal complaints should also be evaluated for *Helicobacter pylori* infection as this organism has been connected to the development of stomach cancer.⁴⁶

Noninfectious Presentations

However, a significant percent of patients with CVID over the years have come to medical attention with one or more of what is considered noninfectious, autoimmune, inflammatory, and/or lymphoproliferative manifestations, as outlined.^{4,47} In a previous European report, 45 years after the diagnosis of CVID was made, only 58% of subjects with this phenotype survived, in comparison to 76% for age-matched men and 85% for age-matched women.⁶ In the United States, the numbers were similar in 2012, with 53% of CVID men and 58% of CVID women in this category surviving, in comparison to 68% and 80% of subjects with the “infection only” phenotype.⁴⁸ The factors identified in large cohorts as being the major contributing factors to poorer prognosis include lymphoma, interstitial lung disease, and chronic enteropathy.^{49,50} This was less clear for patients with autoimmunity and/or granulomatous disease. Splenomegaly alone is not a risk factor, but the presence of a very enlarged spleen may hint at a higher risk in the affected patient group.⁵¹ As these conditions are now a leading cause of concern, the particular focus of research in CVID strives to understand the causes, mechanisms, and treatment of these complications. These conditions are outlined in large cohorts in Table 3,^{6,17,37,52,53} and the current treatment of these is outlined in

Table 4. One poorly understood aspect of CVID is that autoimmune disease is also common and estimated at approximately 25%.^{48,54} The most common of these conditions are immune thrombocytopenic purpura (ITP), autoimmune hemolytic anemia (AIHA), or the combination of both (Evans syndrome); autoimmune neutropenia occurs more rarely.^{55–57} Curiously, ITP and/or AIHA may also be the initial signal of the immune defect in 20% of patients with CVID. For unclear reasons, some of these patients do not have a typical history of recurrent infections, and the acute severe cytopenia may be the first manifestation of the immune defect.

Treatment

IgG Replacement Therapy

The mainstay of treatment in CVID is IgG replacement therapy, by the intravenous or subcutaneous route, chosen on the basis of which therapy is most suitable for the patient. This greatly reduces the incidence of severe bacterial infections such as pneumonia, sepsis, osteomyelitis, and meningitis, as documented in many studies.^{58–60} The individual dose of immunoglobulin (generally 400–600 mg/kg/mo) is determined primarily by measuring IgG serum trough levels, taking into account the baseline level of IgG. In general, trough levels (IgG measured just before a new treatment) of approximately 800 mg/dL are sufficient for most patients,^{58,61,62} but some individuals require higher levels, up to 1000 mg/dL or more, depending on clinical events.⁶³ Higher replacement doses may also be needed especially for subjects with enteropathy, bronchiectasis, and/or splenomegaly.⁶⁴ During follow-up, most physicians check IgG trough levels every 6 months to ensure successful treatment.

Antibiotics and Vaccinations

Antibiotic treatment is also an essential part of the management of patients who have acute or recurrent bacterial infections despite sufficient immunoglobulin replacement. Antibiotic therapy should take into account the usual pathogens that occur in CVID, and culture evidence when possible. The efficiency of antibiotic prophylaxis has been debated but has demonstrated value in a double-blind, placebo-controlled study.⁶⁵ When used, common antibiotics include macrolides (mainly azithromycin 250 mg taken 3 times a week), amoxicillin, or co-trimoxazole-trimethoprim, especially in patients with chronic lower respiratory tract disease.^{65–67} In patients with continued sinus disease, additional measures include topical steroids, mucolytic agents, and saline nasal washes. As immunoglobulin replacement offers passive humoral immunity against vaccine-preventable illnesses such as tetanus, diphtheria, measles, mumps, rubella, pneumococcus, and chicken pox, further immunizations are not required, but may be given as small increases in antibody protection may occur in those with residual function. An annual vaccination against influenza may afford some degree of improved T-cell immunity and may be used. Human papillomavirus vaccines, the RSV vaccine, COVID vaccines, or the inactive recombinant *Herpes zoster* vaccine (Shingrix) can be used with the same schedule as for other subjects, as these contain no infectious components.

Complications and Management

Pulmonary Disease

Repeated respiratory infections, chronic pulmonary disease, including bronchiectasis, interstitial lung disease, and fibrosis, leading to respiratory failure may develop over time in CVID.^{37,68–71} Figure 3A–D reveals the radiologic findings in 4 of these subjects. The pathology reveals lymphocytic interstitial pneumonitis, lymphoid hyperplasia, follicular bronchiolitis, cryptic organizing pneumonia,

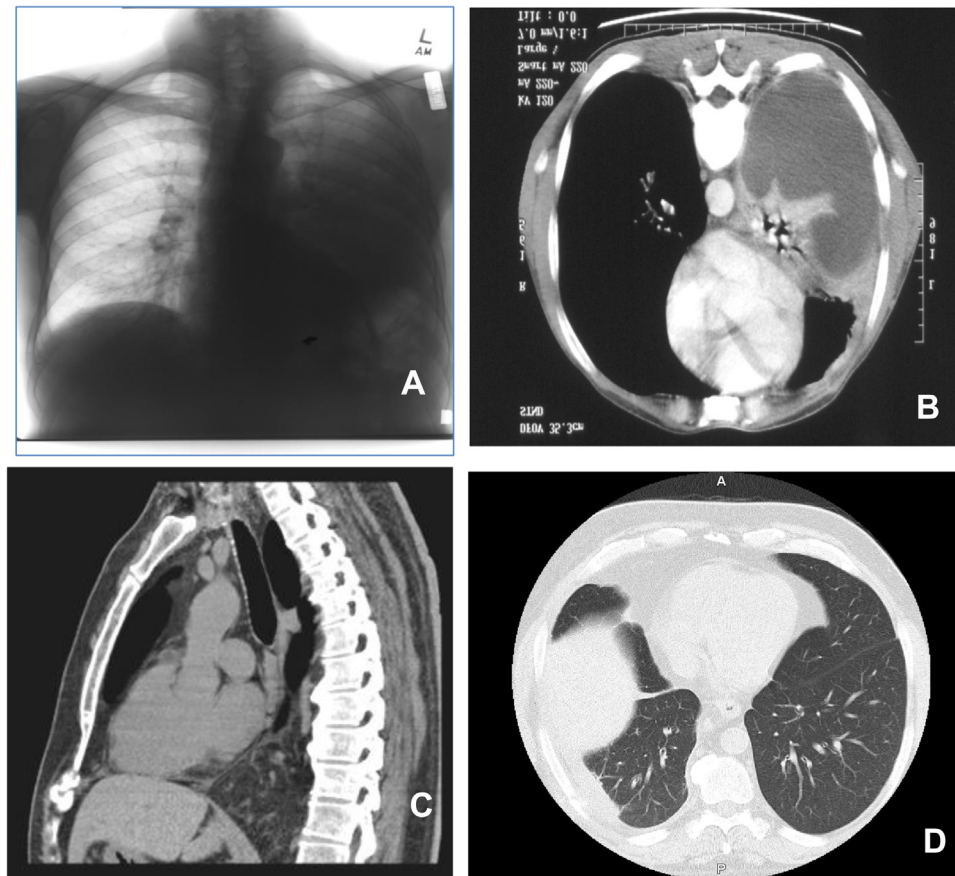


Figure 2. A and B, Chest radiograph of a 40-year-old previously undiagnosed patient with 1 prior episode of pneumonia, now with a second pneumonia, with a large empyema due to *Streptococcus pneumoniae*. C and D, Chest radiograph of a 50-year-old previously undiagnosed patient, with 1 prior episode of hemolytic anemia, now with a large left lower quadrant bacterial pneumonia, also with an empyema in the same region.

granulomatous disease, or combinations of these, including T lymphocytes and B cells forming in what appears to be germinal centers. When granuloma is found on pathology, the term “granulomatous lymphocytic interstitial lung disease” has been used.^{68,69} The uncontrolled, continuous inflammatory changes can lead to alveolar lung damage, pulmonary hypertension, need for chronic oxygen therapy, and ultimately diminished survival.^{68,69,71} Although higher doses of immunoglobulin (600 mg/kg/mo) may help to prevent infections and possibly chronic lung disease,^{72,73} immunoglobulin replacement alone is not likely to improve the lung disease, and other therapies

are needed. When lung disease has developed, bronchoscopy and biopsy are needed to exclude chronic infections and to examine local immunologic changes in the bronchoalveolar tissues. Although systemic steroids can be used for brief periods, other therapies will be needed.⁷⁴ As lung biopsies of patients with ILD often contain a T-cell infiltrate with a variable contribution of B cells forming tertiary structures, combination therapies with rituximab/azathioprine or mycophenylate have been used with success, demonstrating both improved lung functions and radiologic appearance.^{75–77} Although the microbiology of the lungs is usually unrevealing, patients with ILD

Table 3
Noninfectious Complications

Condition	Number	Ho and Cunningham-Rundles ⁵² (n = 623)	Chapel et al ¹⁶ (n = 334)	Quinti et al ³⁷ (n = 224)	Wehr et al ¹⁷	Farmer et al ¹⁵³ (n = 205)
Infection only	199	31.9	26%	NR	NR	NR
Noninfectious complications	424	68.1	74%	NR	NR	NR
Autoimmunity	207	33.2	NR	25.9%	20.3%	NR
Chronic lung disease	189	30.3	NR	46.4%	NR	NR
Lymphoid hyperplasia/splenomegaly	130	20.9	30%	26.4% ^b	40.5% ^b	25.9% ^b
Gastrointestinal disease	108	17.3	9%	22.4% ^c	NR	21.5%
Liver disease	79	12.7	9% ^a	NR	9.3%	9.3%
Granulomas	58	9.3	8%	NR	11.6%	20%
Lymphoma	42	6.7	3%	1.8%	NR	5%
Other malignancies	40	6.7	3% 4.5%	4.5% NR	NR	22% ^d

Abbreviation: NR, not reported.

^aCategorized as hepatomegaly in the original study.

^bCategorized as splenomegaly in the original studies.

^cCategorized as chronic diarrhea in the original study.

^dCategorized as solid organ malignancy in the original study.

Table 4
Treatment and Evaluation of Noninfectious Complications

Complications	Condition	Evaluation/treatment
Autoimmunity	• ITP, AIHA, or neutropenia • Rheumatoid arthritis • Uveitis, vasculitis, alopecia, sicca syndrome, and systemic lupus erythematosus.	• Steroids, higher doses of IVIG, rituximab • Hydroxychloroquine, biologics • As usually treated; hydroxychloroquine, biologics
Chronic lung disease	• Bronchiectasis • Interstitial lung disease • Granulomatous lung disease	• Cultures/antibiotics • Antibiotics/rituximab/azathioprine or mycophenylate, abatacept • Antibiotics/rituximab/azathioprine or mycophenylate, abatacept
Gastrointestinal disease	• Infections • Inflammatory bowel disease • Gastric adenocarcinoma. • Lymphoma and other malignancies	• Culture and PCR/antibiotics as indicated • Budesonide, cholestyramine, 6-mercaptopurine, biologics • Usual treatment/but test for <i>Helicobacter pylori</i> • Usual treatments
Liver disease	• Infections • Nodular regenerative hyperplasia • Gall bladder disease • Biliary cirrhosis • Granulomatous hepatitis	• PCR for viral infections/antiviral if needed • Fibroscans/portal pressures?/beta blocker • As usually treated • As usually treated • Steroids? Rituximab? Azathioprine? Mycophenylate?
Lymphoma	• Lymphoma	• As usually treated • Possible transplant
Other malignancies	• Organ cancers	• As usually treated

Abbreviations: AIHA, autoimmune hemolytic anemia; ITP, immune thrombocytopenic purpura; IVIG, intravenous immunoglobulin; PCR, polymerase chain reaction.

may also benefit from antibiotic prophylaxis, especially with macrolides which provide anti-inflammatory effects.^{65,75,76}

Granulomatous Disease

Granulomatous disease occurs in between 8% and 22% of subjects with CVID, and if diagnosed in advance of the recognition of the

immune defect, it may be mislabeled as sarcoidosis.^{69,78–83} Lungs, lymph nodes, and spleen are the most frequently affected locations, although the skin, liver, bone marrow, kidney, gastrointestinal tract, and/or brain may be involved.^{71,78,84–86} The granuloma in CVID are typically well formed and appears distinct from sarcoidosis by a number of parameters based on clinical, pathologic, and radiologic evidence.^{87,88} Treatment of granulomatous disease depends on the

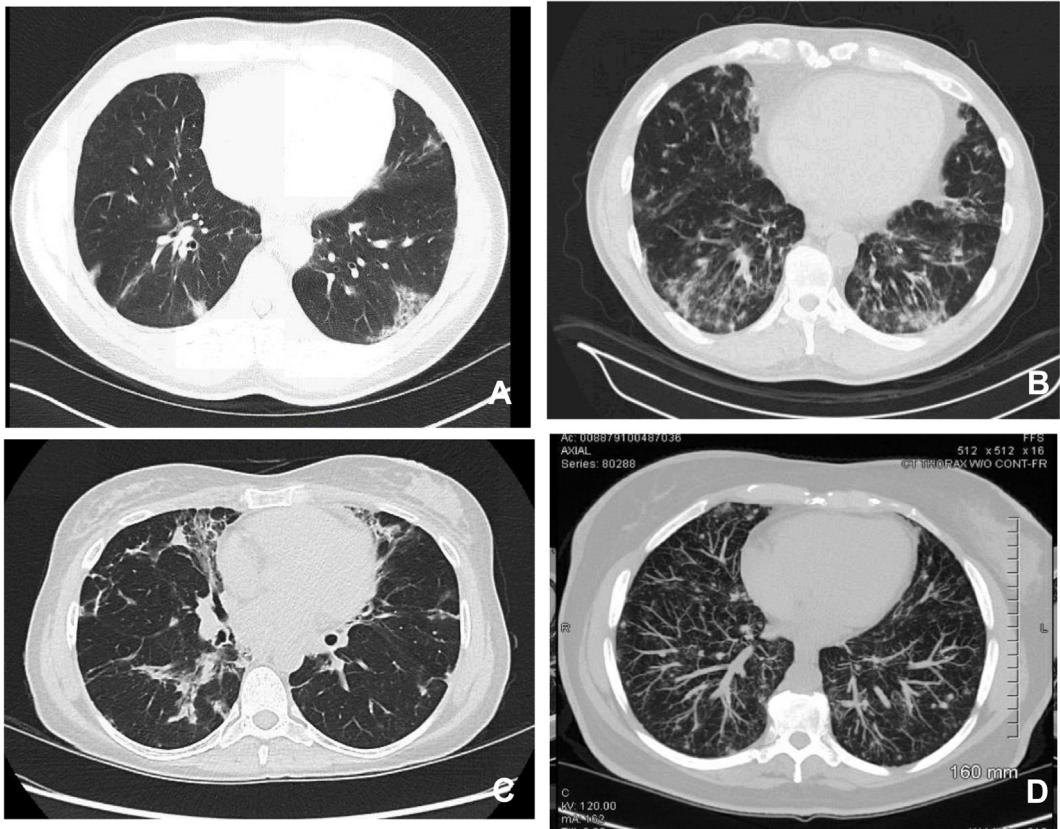


Figure 3. A–D, These are 4 patients with CVID with interstitial lung disease, with biopsy-identified granuloma in B and C (GLILD), but with not in patients A and D. CVID, common variable immunodeficiency; GLILD, granulomatous lymphocytic interstitial lung disease.

site affected. Steroids are often used at the beginning of treatment; when there is pulmonary involvement, rituximab with azathioprine or mycophenylate is currently used.⁷ Other therapies used include anti-Tumor necrosis factor inhibitors and/or abatacept. Granulomatous infiltration of just the lymph nodes or skin does not necessitate treatment, but specific infections leading to this pathology must be excluded.⁸⁹ When granuloma are noted in liver biopsies, short courses of low doses of oral steroids have been most often applied, but the best treatment is currently not known.

Autoimmunity

With lack of antibody production, it may seem improbable that autoimmune disease could occur in CVID, but in fact autoimmunity is common and estimated at approximately 25%.^{48,54} The most common of these conditions is ITP, AIHA, the combination of both (Evans syndrome), or, more rarely, autoimmune neutropenia.^{55–57} ITP and/or AIHA, occurring in 20%, are the most common autoimmune manifestations in CVID and may be the initial signal of the immune defect.⁹⁰ However, other organ/tissue-based autoimmune diseases also occur, including rheumatoid arthritis, pernicious anemia, vitiligo, thyroiditis, enteropathy, hepatitis, less often uveitis, vasculitis, alopecia, primary biliary cirrhosis, sicca syndrome, and very rarely systemic lupus erythematosus. Frequently applied autoantibody tests are generally not useful in CVID, rendering the diagnosis often a clinical or histologic diagnosis. For treating immune cytopenias in CVID, higher doses of immunoglobulin (1 g/kg body weight) can be used to supplement baseline therapy for acute ITP or AIHA; but, often, intravenous steroids (3×100 mg to 1 g methylprednisolone/d) followed by moderate doses of oral steroids tapered in several weeks are necessary for more permanent control. Rituximab (weekly 375 mg/m² for 4 weeks or 1000 mg a week for 2 weeks) has been used in CVID for both refractory and recurrent ITP and/or AIHA with benefit in many patients. Repeat courses of rituximab may also be needed.⁹¹ Splenectomy can be successful even after rituximab failure,^{48,92} but it is associated with an increased risk of severe infections,^{36,93} and its use is discouraged. However, if this has been done, adequate immunoglobulin replacement is important after this procedure, as well as patient education and consideration of antimicrobial prophylaxis. Thrombopoietin-receptor agonists can also be used with benefit in subjects with chronic ITP.⁹⁴ The treatment for organ-specific autoimmune disorders resembles standard therapy but with an emphasis on minimal immune suppression. In general, an infectious cause, for example of mycoplasma leading to a form of arthritis, has to be excluded before implementation of any immunosuppressive therapy for these conditions. Somewhat more confusing is the recommendation for using the antiviral entecavir to prevent rituximab-associated HBV reactivation if antibody to hepatitis B core antigen (anti-HBc) is identified. As this antibody is found in therapeutic immune globulins,⁹⁵ patients with CVID may be instructed to take this antiviral if rituximab is to be used, even with no history or evidence of infection.

Gastrointestinal Disease

Inflammatory bowel disease is a significant chronic problem in 19% to 32% of patients with CVID.^{43,96–98} Gastrointestinal disease includes infections, gastritis, atrophic gastritis, ulcerative colitis, microscopic colitis, Crohn's like-disease, colonic polyps, lymphoid overgrowth, and gastrointestinal malignancies, such as gastric adenocarcinoma, lymphoma, and colon malignancies. For a patient with diarrhea, a first step in the workup is to examine stool by polymerase chain reaction to exclude a known organism which may be treated. Norovirus may however be identified, and no clear treatment has emerged.⁴⁵ For others, a celiac sprue-like enteropathy, when persistent, can lead to chronic diarrhea, weight loss, steatorrhea, and significant

malabsorption, with loss of both minerals and fat-soluble vitamins. On biopsy, the gastrointestinal mucosa of patients reveals villous blunting and often contains increased numbers of intraepithelial T (usually CD8 +) lymphocytes, lymphoid aggregates, and lack of plasma cells.^{42,43,96} The diagnosis of celiac disease is challenging in CVID, as autoantibody detection will not be helpful. As the histopathology cannot reliably discriminate between celiac disease and CVID enteropathy, HLA typing for the relevant markers (HLA-DQ2 and HLA-DQ8) can be helpful, if only to exclude this diagnosis.⁹⁹ Nodular lymphoid hyperplasia is common in any area of the CVID gastrointestinal tract; when massive, this can be associated with chronic diarrhea, weight loss, and possibly intussusception. Bacterial overgrowth appears common, and when documented, it may be ameliorated by broad-spectrum antibiotics (metronidazole or ciprofloxacin or nonresorbable antibiotics). The use of probiotic drugs has not been sufficiently evaluated for general recommendations. *H. pylori* may be associated with gastritis⁴⁶ but otherwise may be asymptomatic; however, if identified, it should be treated as this organism is associated with gastric malignancy. For any patient with upper gastrointestinal symptoms, endoscopy to exclude any pathology is in order. Gastric cancer and small bowel lymphomas can develop in CVID and can be excluded by endoscopy with biopsies as required. For less accessible segments of the bowel, abdominal magnetic resonance imaging may be needed.

For the inflammatory bowel-like disease, non-absorbed oral steroids such as budesonide are often used. Cholestyramine can be used as bulk agent to slow diarrhea. Low-dose corticosteroids such as prednisone can be used in doses of 10 mg/d; however, higher doses lead to a significant risk of infections and, over time, osteoporosis, and can be difficult to taper. There are no guidelines for steroid-sparing agents in the context of enteropathy. Immunomodulators, such as azathioprine or 6-mercaptopurine, can be used safely as the doses used (as for Crohn's disease) are low and do not appear to affect standard T-cell and B-cell function tests.⁴³ The anti- $\alpha 4\beta 7$ integrin (vedolizumab) may be helpful with endoscopic and clinical improvement in some¹⁰⁰ but not in other cases.¹⁰¹ Abatacept and sirolimus may also be used, but clinical guidelines are lacking. Even more targeted therapies, such as adalimumab, infliximab, ustekinumab, and Janus kinase inhibitors are other examples of targeted biologics that can be useful in different cases. Treatment for small bowel enteropathy can be quite difficult in CVID, and a successful approach for one patient may not work in another. In general, co-managing patients with a gastroenterologist who is familiar with this immune defect provides the best patient care.

Liver Disease

Liver disease leading to increases of liver enzyme levels is common in CVID, especially increased serum alkaline phosphatase.⁵⁰ Although hepatitis B and C and CMV infections have to be excluded by polymerase chain reaction, these are uncommon in CVID unless risk factors are present. Gall bladder disease with cholestasis, primary biliary cirrhosis, and immune or granulomatous hepatitis may occur. The most frequently identified liver disease on biopsy is nodular regenerative hyperplasia, which may be indolent, but it may lead to portal hypertension, splenomegaly, esophageal varices, and ascites, but rarely jaundice.^{50,102–105} The cause of this complication remains unclear, but monitoring for increasing liver disease and fibro scans at intervals can be helpful as well as endoscopy to exclude formation of varices. There is no known treatment for this form of CVID liver disease, thus rendering control of secondary complications of portal hypertension is the main goal of therapy. Transjugular intrahepatic portosystemic shunts have been used, but with variable success.^{106,107} Liver transplantation has been performed, but the outcome has not been very encouraging with recurrent disease in the transplanted liver in some.^{108,109}

Lymphoid Hyperplasia, Splenomegaly

Cervical, mediastinal, and abdominal lymphoid hyperplasia are found in at least 20% of subjects with CVID, and 30% may have an enlarged spleen. Lymphoid infiltrates can occur in lung or other organs such as the liver or kidneys. Biopsy results of lymph nodes usually reveal atypical lymphoid hyperplasia, reactive lymphoid hyperplasia, or granulomatous inflammation. Splenomegaly can be massive and can be associated with some degree of hypersplenism and abdominal discomfort. Although splenectomy was previously performed for some cases of ITP or other cytopenias, removal of the spleen is to be avoided, as other measures to reverse the cytopenia are usually successful.

Lymphoma and Other Cancers

In CVID, there appears to be an estimated 1.8- to 5-fold increased risk of developing cancer overall.^{110–113} The most frequently diagnosed cancer is Epstein-Barr virus-negative B-cell lymphoma. Expansion of mucosa-associated lymphoid tissue of the gastrointestinal tract or of the bronchial tissue and diffuse large B-cell lymphoma, not associated with lymph nodes, are the typical entities of non-Hodgkin's lymphoma found in CVID.^{48,52,114,115} In a study of 176 subjects in a European study, the observed-to-expected ratio for lymphoma in CVID was 12.1, and for stomach cancer was 10.3, whereas other forms of solid cancer had no similar increased risk.¹¹³ Lymphoma is a major risk factor for mortality in CVID.^{48,51,115} Gastric cancer has been noted in some series and might be secondary to infection with *H pylori*⁴⁶; with increasing use of antibiotics, this complication appears to be less common. As subjects with CVID can have unusual lymphoid structures with loss of characteristic boundaries, it is important for an experienced pathologist to examine these tissues in close collaboration with the treating clinician.⁵²

Conclusion

Due in great part to replacement immunoglobulin increasing serum IgG levels well into the normal range, the overall course of CVID has changed dramatically in the past 40 years, with greatly enhanced survival. Before this time, many subjects had severe bacterial infections, prolonged hospitalizations, and shortened survival, often due to lung disease. With the introduction of immunoglobulin therapy, recurrent episodes of pneumonia, sepsis, meningitis, osteomyelitis, and so on are quite rare. The additional development of long-acting oral antibiotics has also increased our patients' quality of life. Evaluating records of subjects with CVID treated for most of their lives with appropriate doses of IgG reveals that approximately half of all patients, especially those who present with "infection only," do well, with few unplanned medical encounters. Survival curves constructed at intervals in the past few decades have illustrated this remarkable achievement. As these conditions are now a leading cause of concern, the particular focus of research in CVID strives to understand the causes, mechanisms, and treatment of these complications. Although medical outcomes are clearly improved in the earlier decades, the perceived health of patients, both physical and mental, has been investigated. Evaluating this in 209 patients with CVID, 44% had relatively high physical and mental health status; 21% had low physical and mental health status; whereas the other groups were mixed with high mental and low physical health (15.8%) or vice versa (19.1%). Most interesting was that the distinct clusters of perceived health were largely independent of CVID complications or laboratory parameters reported by their physicians.⁵³ With advances in genetic testing, we have now identified a number of the mechanisms leading to B-cell failure in approximately 30% of patients, but for the rest, the reasons for loss of immune function have not yet been clarified. In a few cases, the genetic information has led to remarkable

therapeutic approaches (as discussed previously). However, in other cases, biologic agents that target more general aspects of inflammation are required, such as steroids, rituximab, azathioprine, mycophenolate, sirolimus, or various newer biologics. Ongoing work in genetics of CVID and better understanding of the activated biologic pathways will hopefully add additional clues to more targeted medical and biological treatments.

Disclosures

Dr Cunningham-Rundles has served as a consultant for Pharming, Grifols and Sanofi; a speaker for Grifols and Takeda; and serves on Data Safety Monitoring Boards for Otsuka and Grifols.

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References

- Bonilla FA, Barlan I, Chapel H, Costa-Carvalho BT, Cunningham-Rundles C, de la Morena MT, et al. International consensus document (ICON): common variable immunodeficiency disorders. *J Allergy Clin Immunol Pract*. 2016;4(1):38–59.
- Seidel MG, Kindle G, Gathmann B, Quinti I, Buckland M, van Montfrans J, et al. The European Society for Immunodeficiencies (ESID) Registry Working Definitions for the Clinical Diagnosis of Inborn Errors of Immunity. *J Allergy Clin Immunol Pract*. 2019;7(6):1763–1770.
- Tangye SG, Al-Herz W, Bousfiha A, Cunningham-Rundles C, Franco JL, Holland SM, et al. Human inborn errors of immunity: 2022 update on the classification from the international union of immunological societies expert committee. *J Clin Immunol*. 2022;42(7):1473–1507.
- Resnick ES, Moshier EL, Godbold JH. Cunningham-Rundles C. Morbidity and mortality in common variable immune deficiency over 4 decades. *Blood*. 2012;119(7):1650–1657.
- Slade CA, Bosco JJ, Binh Giang T, Kruse E, Stirling RG, Cameron PU, et al. Delayed diagnosis and complications of predominantly antibody deficiencies in a cohort of Australian adults. *Front Immunol*. 2018;9:694.
- Chapel H, Lucas M, Lee M, Bjorkander J, Webster D, Grimbacher B, et al. Common variable immunodeficiency disorders: division into distinct clinical phenotypes. *Blood*. 2008;112(2):277–286.
- Remiker A, Bolling K, Verbsky J. Common variable immunodeficiency. *Med Clin North Am*. 2024;108(1):107–121.
- Bruton OC. Agammaglobulinemia. *Pediatrics*. 1952;9(6):722–728.
- Sanford JP, Favour CB, Tribeman MS. Absence of serum gamma globulins in an adult. *N Engl J Med*. 1954;250(24):1027–1029.
- Arends T, Coonrad EV, Rundles RW. Serum proteins in Hodgkin's disease and malignant lymphoma. *Am J Med*. 1954;16(6):833–841.
- Good RA, Varco RL, Aust JB, Zak SJ. Transplantation studies in patients with agammaglobulinemia. *Ann N Y Acad Sci*. 1957;64(5):882–924.
- Wolf HM, Thon V, Litzman J, Eibl MM. Detection of impaired IgG antibody formation facilitates the decision on early immunoglobulin replacement in hypogammaglobulinemic patients. *Front Immunol*. 2015;6:32.
- Orange JS, Ballou M, Stiehm ER, Ballas ZK, Chinen J, De La, Morena M, et al. Use and interpretation of diagnostic vaccination in primary immunodeficiency: a working group report of the Basic and Clinical Immunology Interest Section of the American Academy of Allergy, Asthma & Immunology. *J Allergy Clin Immunol*. 2012;130(3)(suppl):S1–S24.
- Agematsu K, Futatani T, Hokibara S, Kobayashi N, Takamoto M, Tsukada S, et al. Absence of memory B cells in patients with common variable immunodeficiency. *Clin Immunol*. 2002;103(1):34–42.
- Warnatz K, Denz A, Dräke R, Braun M, Groth C, Wolff-Vorbeck G, et al. Severe deficiency of switched memory B cells (CD27(+)/IgM(-)/IgD(-)) in subgroups of patients with common variable immunodeficiency: a new approach to classify a heterogeneous disease. *Blood*. 2002;99(5):1544–1551.
- Piqueras B, Lavenu-Bombled C, Galicier L, Bergeron-van der Cruyssen F, Mouthon L, Chevrete S, et al. Common variable immunodeficiency patient classification based on impaired B cell memory differentiation correlates with clinical aspects. *J Clin Immunol*. 2003;23(5):385–400.
- Wehr C, Kivioja T, Schmitt C, Ferry B, Witte T, Eren E, et al. The EUROclass trial: defining subgroups in common variable immunodeficiency. *Blood*. 2008;111(1):77–85.
- Ochrop ML, Goldacker S, May AM, Rizzi M, Draeger R, Hauschke D, et al. T and B lymphocyte abnormalities in bone marrow biopsies of common variable immunodeficiency. *Blood*. 2011;118(2):309–318.
- Sánchez-Ramón S, Radigan L, Yu JE, Bard S, Cunningham-Rundles C. Memory B cells in common variable immunodeficiency: clinical associations and sex differences. *Clin Immunol*. 2008;128(3):314–321.

20. Mouillot G, Carmagnat M, Gérard L, Garnier JL, Fieschi C, Vince N, et al. B-cell and T-cell phenotypes in CVID patients correlate with the clinical phenotype of the disease. *J Clin Immunol*. 2010;30(5):746–755.
21. Giovannetti A, Pierdominici M, Mazzetta F, Marziali M, Renzi C, Mileo AM, et al. Unravelling the complexity of T cell abnormalities in common variable immunodeficiency. *J Immunol*. 2007;178(6):3932–3943.
22. Gathmann B, Mahlaoui N, CEREDIHGérard L, Oksenhendler E, Warnatz K, et al. Clinical picture and treatment of 2212 patients with common variable immunodeficiency. *J Allergy Clin Immunol*. 2014;134(1):116–126.
23. Maglione PJ, Gyimesi G, Cols M, Radigan L, Ko HM, Weinberger T, et al. BAFF-driven B cell hyperplasia underlies lung disease in common variable immunodeficiency. *JCI Insight*. 2019;4(5):e122728.
24. Jolles S. The variable in common variable immunodeficiency: a disease of complex phenotypes. *J Allergy Clin Immunol Pract*. 2013;1(6):545–556. quiz 57.
25. Wiseman DH, May A, Jolles S, Connor P, Powell C, Heeney MM, et al. A novel syndrome of congenital sideroblastic anemia, B-cell immunodeficiency, periodic fevers, and developmental delay (SIFD). *Blood*. 2013;122(1):112–123.
26. Peng XP, Caballero-Oteyza A, Grimbacher B. Common variable immunodeficiency: more pathways than roads to Rome. *Annu Rev Pathol*. 2023;18:283–310.
27. Abolhassani H, Hammarstrom L, Cunningham-Rundles C. Current genetic landscape in common variable immune deficiency. *Blood*. 2020;135(9):656–667.
28. Rojas-Restrepo J, Caballero-Oteyza A, Huebscher K, Haberstroh H, Fliegau M, Keller B, et al. Establishing the molecular diagnoses in a cohort of 291 patients with predominantly antibody deficiency by targeted next-generation sequencing: experience from a monocentric study. *Front Immunol*. 2021;12:786516.
29. Maffucci P, Filion CA, Boisson B, Itan Y, Shang L, Casanova JL, et al. Genetic diagnosis using whole exome sequencing in common variable immunodeficiency. *Front Immunol*. 2016;7:220.
30. Tuijnenburg P, Lango Allen H, Burns SO, Greene D, Jansen MH, Staples E, et al. Loss-of-function nuclear factor κ B subunit 1 (NFKB1) variants are the most common monogenic cause of common variable immunodeficiency in Europeans. *J Allergy Clin Immunol*. 2018;142(4):1285–1296.
31. Cunningham-Rundles C, Casanova JL, Boisson B. Genetics and clinical phenotypes in common variable immunodeficiency. *Front Genet*. 2023;14:1272912.
32. von Spee-Mayer C, Echternach C, Agarwal P, Gutenberger S, Soetedjo V, Goldacker S, et al. Abatacept use is associated with steroid dose reduction and improvement in fatigue and CD4-dysregulation in CVID patients with interstitial lung disease. *J Allergy Clin Immunol Pract*. 2021;9(2):760–770e10.
33. Rao VK, Webster S, Sedivá A, Plebani A, Schuetz C, Shcherbina A, et al. A randomized, placebo-controlled phase 3 trial of the PI3K δ inhibitor leniolisib for activated PI3K δ syndrome. *Blood*. 2023;141(9):971–983.
34. Touw CM, van de Ven AA, de Jong PA, Terheggen-Lagro S, Beek E, Sanders EA, et al. Detection of pulmonary complications in common variable immunodeficiency. *Pediatr Allergy Immunol*. 2010;21(5):793–805.
35. Kainulainen L, Nikoskelainen J, Ruuskanen O. Diagnostic findings in 95 Finnish patients with common variable immunodeficiency. *J Clin Immunol*. 2001;21(2):145–149.
36. Cunningham-Rundles C, Bodian C. Common variable immunodeficiency: clinical and immunological features of 248 patients. *Clin Immunol*. 1999;92(1):34–48.
37. Quinti I, Soresina A, Spadaro G, Martino S, Donnanno S, Agostini C, et al. Long-term follow-up and outcome of a large cohort of patients with common variable immunodeficiency. *J Clin Immunol*. 2007;27(3):308–316.
38. Malphettes M, Oksenhendler E, Galicier L, Fieschi C. Granulomatous disease in common variable immunodeficiency. *Rev Med Intern*. 2008;29(1):28–32.
39. Malphettes M, Gérard L, Carmagnat M, Mouillot G, Vince N, Boutboul D, et al. Late-onset combined immune deficiency: a subset of common variable immunodeficiency with severe T cell defect. *Clin Infect Dis*. 2009;49(9):1329–1338.
40. von Spee-Mayer C, Koemm V, Wehr C, Goldacker S, Kindle G, Bulashevskaya A, et al. Evaluating laboratory criteria for combined immunodeficiency in adult patients diagnosed with common variable immunodeficiency. *Clin Immunol*. 2019;203:59–62.
41. Washington K, Stenzel TT, Buckley RH, Gottfried MR. Gastrointestinal pathology in patients with common variable immunodeficiency and X-linked agammaglobulinemia. *Am J Surg Pathol*. 1996;20(10):1240–1252.
42. Daniels JA, Lederman HM, Maitra A, Montgomery EA. Gastrointestinal tract pathology in patients with common variable immunodeficiency (CVID): a clinicopathologic study and review. *Am J Surg Pathol*. 2007;31(12):1800–1812.
43. Agarwal S, Mayer L. Pathogenesis and treatment of gastrointestinal disease in antibody deficiency syndromes. *J Allergy Clin Immunol*. 2009;124(4):658–664.
44. Oksenhendler E, Gérard L, Fieschi C, Malphettes M, Mouillot G, Jaussaud R, et al. Infections in 252 patients with common variable immunodeficiency. *Clin Infect Dis*. 2008;46(10):1547–1554.
45. Strohmeier V, Andrieux G, Unger S, Pascual-Reguant A, Klocperk A, Seidl M, et al. Interferon-driven immune dysregulation in common variable immunodeficiency-associated villous atrophy and Norovirus infection. *J Clin Immunol*. 2023;43(2):371–390.
46. Zullo A, Romiti A, Rinaldi V, Vecchione A, Tomao S, Aiuti F, et al. Gastric pathology in patients with common variable immunodeficiency. *Gut*. 1999;45(1):77–81.
47. Pensieri MV, Pulvirenti F, Schieppatti A, Maimaris S, Lattanzio S, Quinti I, et al. The high mortality of patients with common variable immunodeficiency and small bowel villous atrophy. *Scand J Gastroenterol*. 2019;54(2):164–168.
48. Bez P, Warnatz K. Liver disease in primary antibody deficiencies. *Curr Opin Allergy Clin Immunol*. 2024;24(6):434–439.
49. Chapel H, Lucas M, Patel S, Lee M, Cunningham-Rundles C, Resnick E, et al. Confirmation and improvement of criteria for clinical phenotyping in common variable immunodeficiency disorders in replicate cohorts. *J Allergy Clin Immunol*. 2012;130(5):1197–1198.e9.
50. Boileau J, Mouillot G, Gerard L, Carmagnat M, Rabian C, Oksenhendler E, et al. Autoimmunity in common variable immunodeficiency: correlation with lymphocyte phenotype in the French DEFI study. *J Autoimmun*. 2011;36(1):25–32.
51. Cunningham-Rundles C. Hematologic complications of primary immune deficiencies. *Blood Rev*. 2002;16(1):61–64.
52. Ho HE, Cunningham-Rundles C. Non-infectious complications of common variable immunodeficiency: updated clinical spectrum, sequelae, and insights to pathogenesis. *Front Immunol*. 2020;11:149.
53. Farmer JR, Ong MS, Barmettler S, Yonker LM, Fuleihan R, Sullivan KE, et al. Common variable immunodeficiency non-infectious disease endotypes redefined using unbiased network clustering in large electronic datasets. *Front Immunol*. 2017;8:1740.
54. Michel M, Chanet V, Galicier L, Ruivard M, Levy Y, Hermine O, et al. Autoimmune thrombocytopenic purpura and common variable immunodeficiency: analysis of 21 cases and review of the literature. *Medicine (Baltimore)*. 2004;83(4):254–263.
55. Wang J, Cunningham-Rundles C. Treatment and outcome of autoimmune hematologic disease in common variable immunodeficiency (CVID). *J Autoimmun*. 2005;25(1):57–62.
56. Orange JS, Hossny EM, Weiler CR, Ballow M, Berger M, Bonilla FA, et al. Use of intravenous immunoglobulin in human disease: a review of evidence by members of the Primary Immunodeficiency Committee of the American Academy of Allergy, Asthma and Immunology. *J Allergy Clin Immunol*. 2006;117(4)(suppl):S525–S553.
57. Busse PJ, Razvi S, Cunningham-Rundles C. Efficacy of intravenous immunoglobulin in the prevention of pneumonia in patients with common variable immunodeficiency. *J Allergy Clin Immunol*. 2002;109(6):1001–1004.
58. Nolte MT, Pirofsky B, Gerritz GA, Golding B. Intravenous immunoglobulin therapy for antibody deficiency. *Clin Exp Immunol*. 1979;36(2):237–243.
59. Roifman CM, Berger M, Notarangelo LD. Management of primary antibody deficiency with replacement therapy: summary of guidelines. *Immunol Allergy Clin North Am*. 2008;28(4):875–876. x.
60. Shehata N, Palda V, Bowen T, Haddad E, Issekutz TB, Mazer B, et al. The use of immunoglobulin therapy for patients with primary immune deficiency: an evidence-based practice guideline. *Transfus Med Rev*. 2010;24(suppl 1):S28–S50.
61. Orange JS, Grossman WJ, Navickis RJ, Wilkes MM. Impact of trough IgG on pneumonia incidence in primary immunodeficiency: A meta-analysis of clinical studies. *Clin Immunol*. 2010;137(1):21–30.
62. Lucas M, Hugh-Jones K, Welby A, Misbah S, Spaeth P, Chapel H. Immunomodulatory therapy to achieve maximum efficacy: doses, monitoring, compliance, and self-infusion at home. *J Clin Immunol*. 2010;30(suppl 1):S84–S89.
63. Miloto C, Pulvirenti F, Cinetto F, Lougaris V, Soresina A, Pecoraro A, et al. Double-blind, placebo-controlled, randomized trial on low-dose azithromycin prophylaxis in patients with primary antibody deficiencies. *J Allergy Clin Immunol*. 2019;144(2):584–593.e7.
64. Sperlich JM, Grimbacher B, Workman S, Haque T, Seneviratne SL, Burns SO, et al. Respiratory infections and antibiotic usage in common variable immunodeficiency. *J Allergy Clin Immunol Pract*. 2018;6(1):159–168.e3.
65. Bates CA, Ellison MC, Lynch DA, Cool CD, Brown KK, Routes JM. Granulomatous-lymphocytic lung disease shortens survival in common variable immunodeficiency. *J Allergy Clin Immunol*. 2004;114(2):415–421.
66. Morimoto Y, Routes JM. Granulomatous disease in common variable immunodeficiency. *Curr Allergy Asthma Rep*. 2005;5(5):370–375.
67. van de Ven AA, de Jong PA, Hoytema van Konijnenburg DP, Kessels OA, Boes M, Sanders EA, et al. Airway and interstitial lung disease are distinct entities in paediatric common variable immunodeficiency. *Clin Exp Immunol*. 2011;165(2):235–242.
68. Bantalib HM, van de Ven A, Jacob J, Daviden JR, Fevang B, Hanitsch LG, et al. Diagnostic testing for interstitial lung disease in common variable immunodeficiency: a systematic review. *Front Immunol*. 2023;14:1190235.
69. Roifman CM, Levison H, Gelfand EW. High-dose versus low-dose intravenous immunoglobulin in hypogammaglobulinemia and chronic lung disease. *Lancet*. 1987;1(8541):1075–1077.
70. Eijkhout HW, van Der Meer JW, Kallenberg CG, Weening RS, van Dissel JT, Sanders LA, et al. The effect of two different dosages of intravenous immunoglobulin on the incidence of recurrent infections in patients with primary hypogammaglobulinemia. A randomized, double-blind, multicenter crossover trial. *Ann Intern Med*. 2001;135(3):165–174.
71. Verbsky JW, Hintermeyer MK, Simpson PM, Feng M, Barbeau J, Rao N, et al. Rituximab and antimetabolite treatment of granulomatous and lymphocytic interstitial lung disease in common variable immunodeficiency. *J Allergy Clin Immunol*. 2021;147(2):704–712.e17.
72. Chase NM, Verbsky JW, Hintermeyer MK, Waukau JK, Tomita-Mitchell A, Casper JT, et al. Use of combination chemotherapy for treatment of granulomatous and lymphocytic interstitial lung disease (GLILD) in patients with common variable immunodeficiency (CVID). *J Clin Immunol*. 2013;33(1):30–39.
73. Maglione PJ, Overbey JR, Cunningham-Rundles C. Progression of common variable immunodeficiency interstitial lung disease accompanies distinct pulmonary and laboratory findings. *J Allergy Clin Immunol Pract*. 2015;3(6):941–950.
74. van de Ven AAJM, Alfaro TM, Robinson A, Baumann U, Bergeron A, Burns SO, et al. Managing granulomatous-lymphocytic interstitial lung disease in common variable immunodeficiency disorders: e-GLILDnet International Clinicians survey. *Front Immunol*. 2020;11:606333.
75. López-Boado YS, Rubin BK. Macrolides as immunomodulatory medications for the therapy of chronic lung diseases. *Curr Opin Pharmacol*. 2008;8(3):286–291.
76. Fasano MB, Sullivan KE, Sarpong SB, Wood RA, Jones SM, Johns CJ, et al. Sarcoidosis and common variable immunodeficiency. Report of 8 cases and review of the literature. *Medicine (Baltimore)*. 1996;75(5):251–261.

77. Mullighan CG, Fanning GC, Chapel HM, Welsh KL. TNF and lymphotoxin- α polymorphisms associated with common variable immunodeficiency: role in the pathogenesis of granulomatous disease. *J Immunol*. 1997;159(12):6236–6241.
78. Cunningham-Rundles C. Common variable immunodeficiency. *Curr Allergy Asthma Rep*. 2001;1(5):421–429.
79. Knight AK, Cunningham-Rundles C. Inflammatory and autoimmune complications of common variable immune deficiency. *Autoimmun Rev*. 2006;5(2):156–159.
80. Ardeniz O, Cunningham-Rundles C. Granulomatous disease in common variable immunodeficiency. *Clin Immunol*. 2009;133(2):198–207.
81. Park JH, Levinson AL. Granulomatous-lymphocytic interstitial lung disease (GLILD) in common variable immunodeficiency (CVID). *Clin Immunol*. 2010;134(2):97–103.
82. Misbah SA, Spickett GP, Esiri MM, Hughes JT, Matthews WB, Thompson RA, et al. Recurrent intra-cranial granulomata presenting as space-occupying lesions in a patient with common variable immunodeficiency. *Postgrad Med J*. 1992;68(799):359–362.
83. Mechanic LJ, Dikman S, Cunningham-Rundles C. Granulomatous disease in common variable immunodeficiency. *Ann Intern Med*. 1997;127(8 Pt 1):613–617.
84. Lin JH, Liebhaver M, Roberts RL, Dyer Z, Stiehm ER. Etanercept treatment of cutaneous granulomas in common variable immunodeficiency. *J Allergy Clin Immunol*. 2006;117(4):878–882.
85. Bouvry D, Mouthon L, Brillet PY, Kambouchner M, Ducroix JP, Cottin V, et al. Granulomatosis-associated common variable immunodeficiency disorder: a case-control study versus sarcoidosis. *Eur Respir J*. 2013;41(1):115–122.
86. van Stigt AC, von der Thüsen JH, Mustafa DAM, van den Bosch TPP, Lila KA, Vадgama D, et al. Granulomas in common variable immunodeficiency display different histopathological features compared to other granulomatous diseases. *J Clin Immunol*. 2024;45(1):22.
87. Rotella K, Schiano TD, Fiel MI, Ho HE, Cunningham-Rundles C. Four-year-history of recurrent fever, skin lesions, and liver abscesses in a patient with common variable immune deficiency due to *Helicobacter cinaedi* infection. *J Clin Immunol*. 2023;44(1):16.
88. Cunningham-Rundles C, Casanova JL, Boisson B. Common variable immunodeficiency: autoimmune cytopenias and advances in molecular diagnosis. *Hematology Am Soc Hematol Educ Program*. 2024;2024(1):137–142.
89. Gobert D, Bussel JB, Cunningham-Rundles C, Galicier L, Dechartres A, Berezné A, et al. Efficacy and safety of rituximab in common variable immunodeficiency-associated immune cytopenias: a retrospective multicentre study on 33 patients. *Br J Haematol*. 2011;155(4):498–508.
90. Wong GK, Goldacker S, Winterhalter C, Grimbacher B, Chapel H, Lucas M, et al. Outcomes of splenectomy in patients with common variable immunodeficiency (CVID): a survey of 45 patients. *Clin Exp Immunol*. 2013;172(1):63–72.
91. Seve P, Bourdillon L, Sarrot-Reynauld F, Ruivard M, Jaussaud R, Bouhour D, et al. Autoimmune hemolytic anemia and common variable immunodeficiency: a case-control study of 18 patients. *Medicine (Baltimore)*. 2008;87(3):177–184.
92. Fevang B. Treatment of inflammatory complications in common variable immunodeficiency (CVID): current concepts and future perspectives. *Expert Rev Clin Immunol*. 2023;19(6):627–638.
93. Victoria L, Maharjan AS, Kostka J, Assenso-Bediako R, Merkert W, Chirch L, et al. Prevalence of hepatitis B core antibody in intravenous immunoglobulin products by chemiluminescent microparticle immunoassay. *J Clin Transl Hepatol*. 2025;13(4):358–360.
94. Malamut G, Verkarre V, Suarez F, Viallard JF, Lascaux AS, Cosnes J, et al. The enteropathy associated with common variable immunodeficiency: the delineated frontiers with celiac disease. *Am J Gastroenterol*. 2010;105(10):2262–2275.
95. Pikkarainen S, Martelius T, Ristimäki A, Siitonen S, Seppänen MRJ, Färkkilä M. A high prevalence of gastrointestinal manifestations in common variable immunodeficiency. *Am J Gastroenterol*. 2019;114(4):648–655.
96. Uzzan M, Ko HM, Mehandru S, Cunningham-Rundles C. Gastrointestinal disorders associated with common variable immune deficiency (CVID) and chronic granulomatous disease (CGD). *Curr Gastroenterol Rep*. 2016;18(4):17.
97. Venhoff N, Emmerich F, Neagu M, Salzer U, Koehn C, Driever S, et al. The role of HLA DQ2 and DQ8 in dissecting celiac-like disease in common variable immunodeficiency. *J Clin Immunol*. 2013;33(5):909–916.
98. Boland BS, Riedl MA, Valasek MA, Crowe SE, Sandborn WJ. Vedolizumab in patients with common variable immune deficiency and gut inflammation. *Am J Gastroenterol*. 2017;112(10):1621.
99. Sifers T, Hirtten R, Mehandru S, Ko HM, Colombel JF, Cunningham-Rundles C. Vedolizumab therapy in common variable immune deficiency associated enteropathy: a case series. *Clin Immunol*. 2020;212:108362.
100. Malamut G, Ziol M, Suarez F, Beaugrand M, Viallard JF, Lascaux AS, et al. Nodular regenerative hyperplasia: the main liver disease in patients with primary hypogammaglobulinemia and hepatic abnormalities. *J Hepatol*. 2008;48(1):74–82.
101. Fuss IJ, Friend J, Yang Z, He JP, Hooda L, Boyer J, et al. Nodular regenerative hyperplasia in common variable immunodeficiency. *J Clin Immunol*. 2013;33(4):748–758.
102. Ward C, Lucas M, Piris J, Collier J, Chapel H. Abnormal liver function in common variable immunodeficiency disorders due to nodular regenerative hyperplasia. *Clin Exp Immunol*. 2008;153(3):331–337.
103. Song J, Lleo A, Yang GX, Zhang W, Bowlus CL, Gershwin ME, et al. Common variable immunodeficiency and liver involvement. *Clin Rev Allergy Immunol*. 2018;55(3):340–351.
104. Regnault D, d'Alterroche L, Nicolas C, Dujardin F, Ayoub J, Perarnau JM. Ten-year experience of transjugular intrahepatic portosystemic shunt for noncirrhotic portal hypertension. *Eur J Gastroenterol Hepatol*. 2018;30(5):557–562.
105. Globig AM, Heeg M, Larsen CS, Ferreira RD, Kindle G, Goldacker S, et al. International multicenter experience of transjugular intrahepatic portosystemic shunt implantation in patients with common variable immunodeficiency. *J Allergy Clin Immunol Pract*. 2021;9(7):2931–2935.e1.
106. Azzu V, Kennard L, Morillo-Gutierrez B, Slatter M, Edgar JDM, Kumararatne DS, et al. Liver disease predicts mortality in patients with X-linked immunodeficiency with hyper-IgM but can be prevented by early hematopoietic stem cell transplantation. *J Allergy Clin Immunol*. 2018;141(1):405–408.e7.
107. Jørgensen SF, Macpherson ME, Bjørø K, Karlens TH, Reims HM, Grzyb K, et al. Liver transplantation in patients with primary antibody deficiency. *J Allergy Clin Immunol*. 2017;139(5):1708–1710.e2.
108. Mayor PC, Eng KH, Singel KL, Abrams SI, Odunsi K, Moysich KB, et al. Cancer in primary immunodeficiency diseases: cancer incidence in the United States Immune Deficiency Network Registry. *J Allergy Clin Immunol*. 2018;141(3):1028–1035.
109. Spector BD, Perry 3rd GS, Gajl-Peczalska KJ, Coccia P, Nesbit ME, Kersey JH. Malignancy in children with and without genetically-determined immunodeficiencies. *Birth Defects Orig Artic Ser*. 1978;14(6A):85–89.
110. Mortaz E, Tabarsi P, Mansouri D, Khosravi A, Garssen J, Velayati A, et al. Cancers related to immunodeficiencies: update and perspectives. *Front Immunol*. 2016;7:365.
111. Mellemkjaer L, Hammarstrom L, Andersen V, Yuen J, Heilmann C, Barington T, et al. Cancer risk among patients with IgA deficiency or common variable immunodeficiency and their relatives: a combined Danish and Swedish study. *Clin Exp Immunol*. 2002;130(3):495–500.
112. Cunningham-Rundles C, Cooper DL, Duffy TP, Strauchen J. Lymphomas of mucosal-associated lymphoid tissue in common variable immunodeficiency. *Am J Hematol*. 2002;69(3):171–178.
113. Quinti I, Agostini C, Tabolli S, Brunetti G, Cinetto F, Pecoraro A, et al. Malignancies are the major cause of death in patients with adult onset common variable immunodeficiency. *Blood*. 2012;120(9):1953–1954.
114. Smith T, Cunningham-Rundles C. Lymphoid malignancy in common variable immunodeficiency in a single-center cohort. *Eur J Haematol*. 2021;107(5):503–516.
115. Bayrhuber M, Tinsel I, Goldacker S, Kindle G, Warnatz K, Farin E, et al. Perceived health of patients with common variable immunodeficiency - a cluster analysis. *Clin Exp Immunol*. 2019;196(1):76–85.